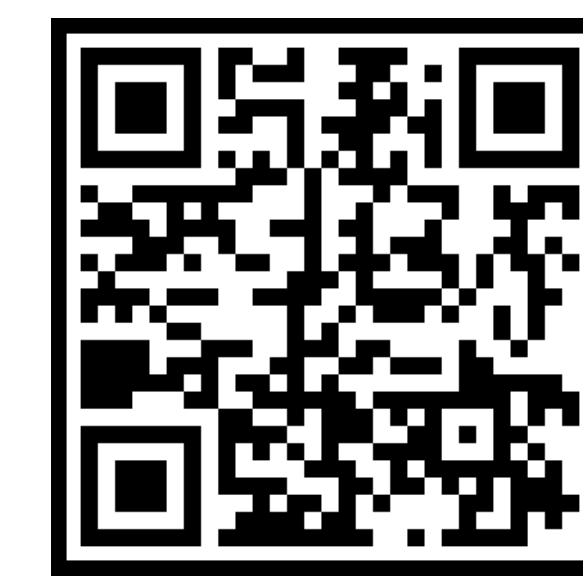


Consistent Scene Understanding in 3D Gaussian Splatting via Multi-Cue Mask Refinement

Hyunjoon Park, Donghyeon Cho

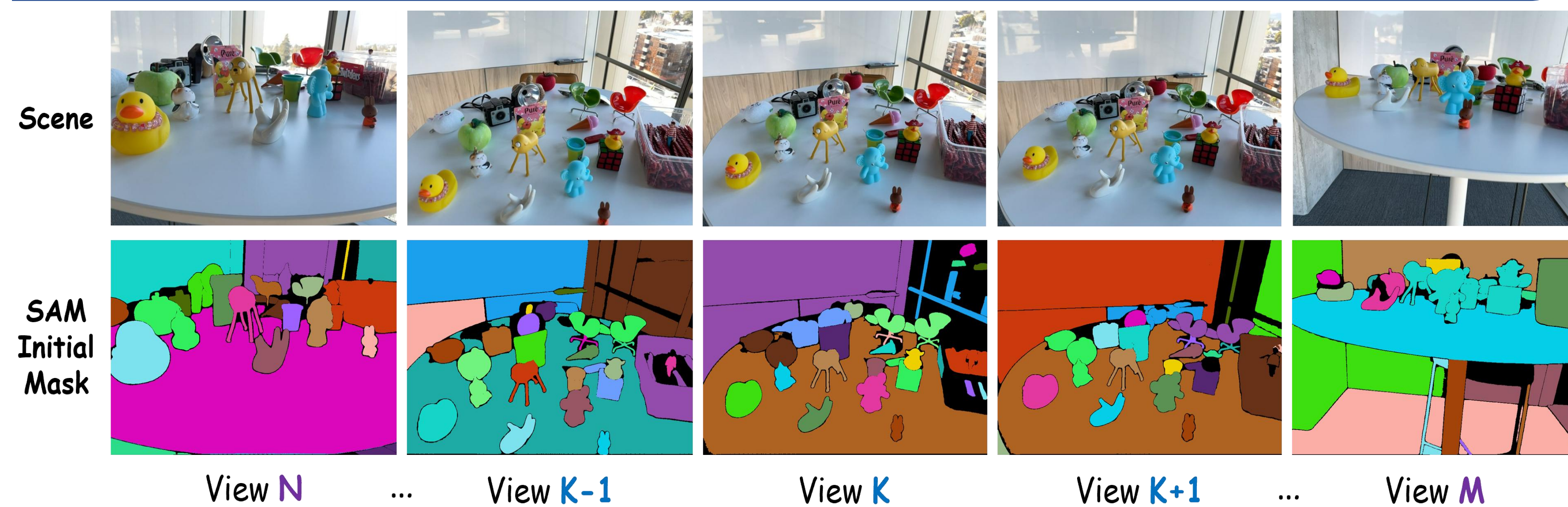
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Project Page



INTRODUCTION



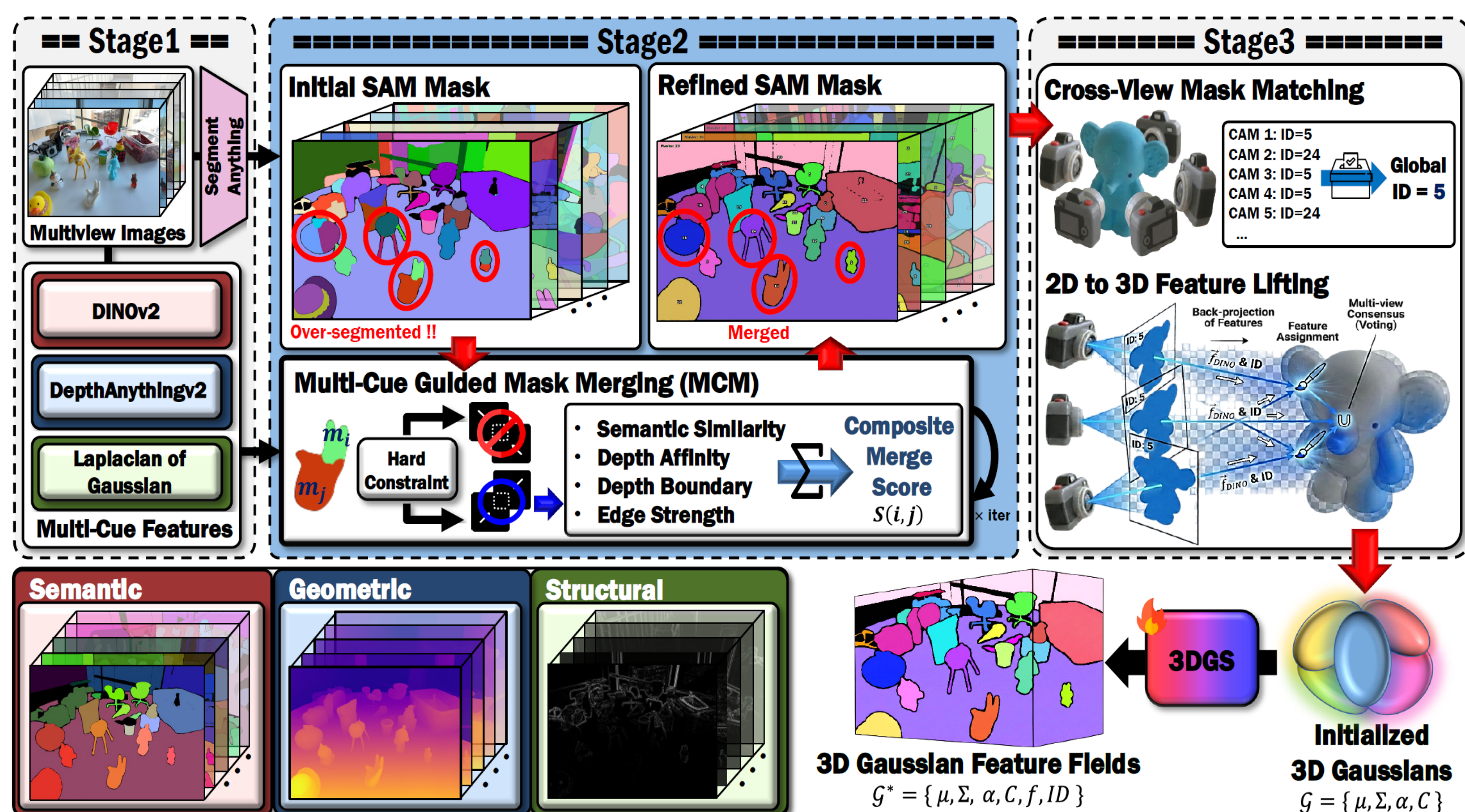
Motivation

- **Cross-view Inconsistency**
- **Over-segmentation of SAM automatic mask generation**
 - Lack 3D spatial awareness and geometric context
 - Merge & Split segmentation mask of a single object erroneously

Contribution

- **Multi-Cue Mask Refinement framework**
 - : Refine SAM-induced over-segmented masks
- **Cross-view Mask Matching**
 - : Establishes globally consistent object identities and suppresses view-dependent label inconsistencies
- **Multiview Consensus Lifting**
- **Performance Enhancement**

METHOD



Stage1

Extract initial SAM mask and multi-cue features, which are semantics (DINOv2), geometric (DepthAnythingv2), and structural (Laplacian of Gaussian Edge Filter).

Stage2

Refine initial SAM mask by performing score-based Multi-Cue-Guided Mask Merging (MCM) algorithm

$$S(i, j) = w_{sem} \cdot \cos(f_i, f_j) + w_{depth} \cdot \exp\left(-\frac{|\bar{d}_i - \bar{d}_j|^2}{2\sigma_d^2}\right) - w_{dbound} \cdot \max(\nabla_D \cap \partial m_{ij}) - w_{edge} \cdot \max(\nabla_{edge} \cap \partial m_{ij})$$

Stage3

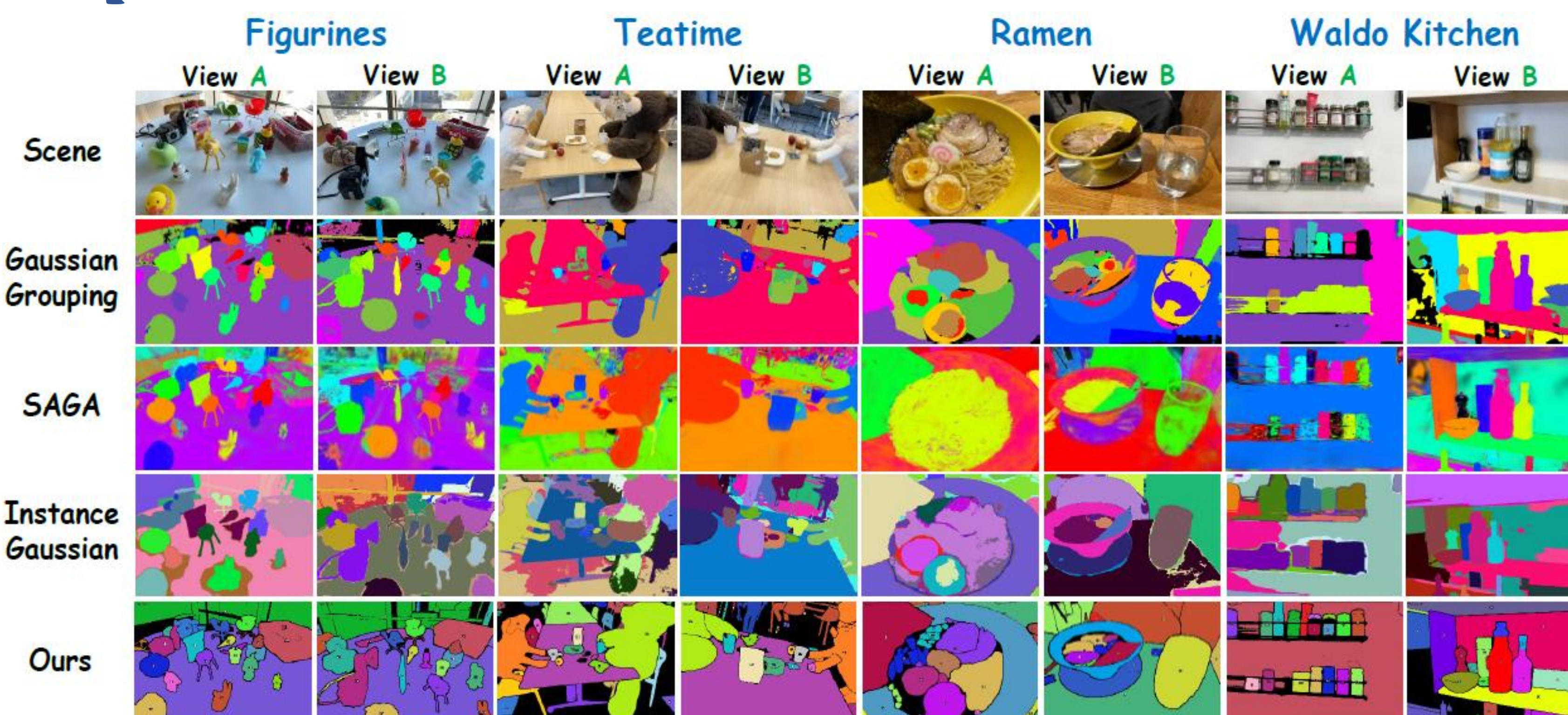
After refinement, to unify view-independent mask IDs into globally consistent ones, Cross-View Mask Matching is performed, and then the 2D features are lifted to 3D space.

CONCLUSION

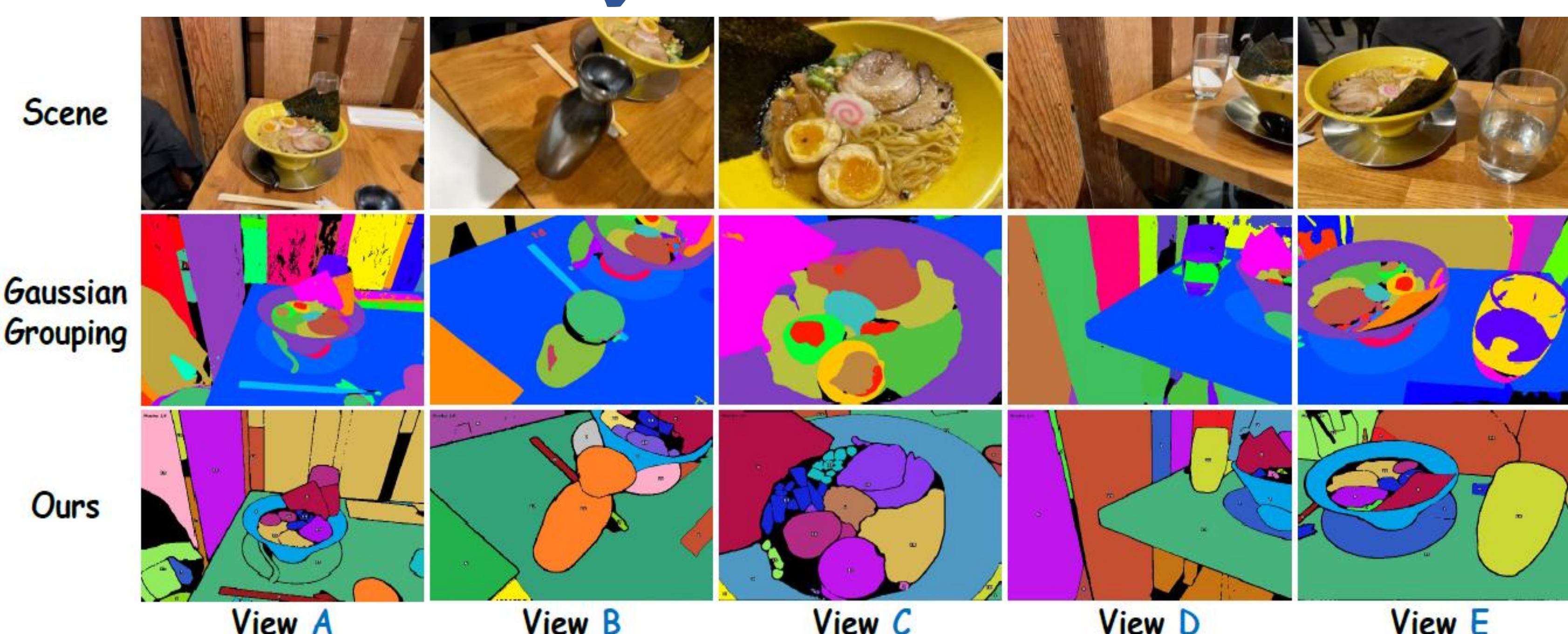
- Propose unified framework for 3D scene decomposition that transforms 2D priors into structured and view-consistent object primitives
- Address over-segmentation of SAM, while ensuring global identity coherence
- Provide robust foundation for structured 3D scene understanding and downstream applications

EXPERIMENTS

Qualitative Results



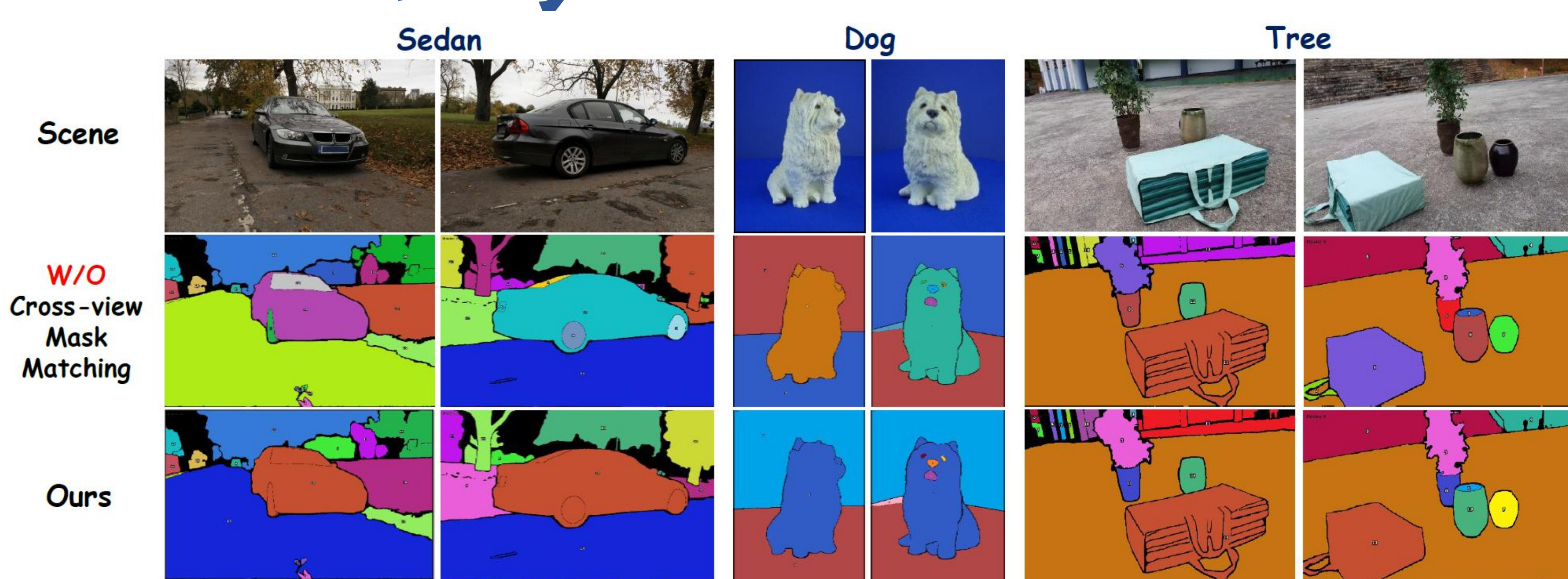
View Consistency



Quantitative Results

Category	Model	PSNR ↑	SSIM ↑	LPIPS ↓	mIoU ↑	mBIoU ↑	#Mask ↓
Standard	3DGS [7]	28.6	0.88	0.11	-	-	-
3D Segmentation	Gau-Group [32]	28.6	0.88	0.12	0.685	0.642	9,627
	SAGA [2]	28.2	0.87	0.15	0.712	0.658	178
	InstanceGaussian [15]	28.8	0.87	0.15	0.702	0.662	101
Feature Field	Feature3DGS [37]	27.5	0.84	0.18	0.538	0.485	-
	GARField [9]	28.1	0.87	0.16	0.710	0.615	-
	CF3 [14]	28.3	0.87	0.14	0.524	0.582	-
Proposed	Ours	28.6	0.89	0.13	0.728	0.677	67

Ablation Study



Configuration	Signals			Metrics			
	DINOv2	LoG	DAV2	mIoU ↑	# Masks ↓	PSNR ↑	SSIM ↑ LPIPS ↓
Baseline [32]	-	-	-	0.685	9,627	28.6	0.88 0.13
Ours (1)	✓	-	-	0.721	245	28.4	0.89 0.14
Ours (2)	✓	✓	-	0.729	112	28.5	0.91 0.12
Ours (Full)	✓	✓	✓	0.741	67	28.6	0.91 0.12

* DAV2 denotes DepthAnythingV2.